KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 2 — THE HIDDEN LAND OPPORTUNITY

Where is Bangladesh's untapped agricultural potential hiding?

Millions of hectares of farmland remain unused or underused across Bangladesh.

Analysis : 64 districts, 5 land types, 22 data points per district
Tools : SQL, Power BI
Dataset : land_data.csv (64 rows, 22 columns)

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THE HIDDEN LAND OPPORTUNITY

THE QUESTION

Where is Bangladesh's untapped agricultural potential hiding?

- This analysis reveals the hidden agricultural capacity Bangladesh is not utilizing.

APPROACH

- Mapped all 64 districts for agricultural land inventory
- Identified key types of underutilized land: fallow, char, hilly, saline, and rice-fallow
- Estimated total untapped agricultural potential (hectare level)
- Analyzed cropping intensity across divisions
- Assessed land loss indicators (urbanization impact)
- Ranked districts based on opportunity size

DATASET OVERVIEW

File : lands.csv
Rows : 64 (one per district)
Columns : 23

Key Fields :

* Agricultural land distribution (total, fallow, char, hilly, saline, haor)
* Untapped production potential (hectare)
* Cropping intensity (%)
* Land loss trends
* Regional classification (division, area type)

Source : SRDI, BBS Agricultural Census, DAE

SQL Queries

Question
Which districts and area types have the lowest cropping intensity, indicating the greatest opportunity to increase food production through better land use?

```
SELECT
    district_name,
    division,
    primary_area_type,

    ROUND(total_agricultural_land_hectare, 0) AS total_land,
    ROUND(total_untapped_potential_hectare, 0) AS untapped_land,

    ROUND(cropping_intensity_pct, 2) AS cropping_intensity_pct

FROM dbo.lands

ORDER BY cropping_intensity_pct ASC;
```

Key Insight:

Cropping intensity across Bangladesh is already high, with most districts clustered around 200%, indicating near full utilization of agricultural cycles.

The small variation between districts suggests that the remaining gap is not about increasing cropping frequency, but improving productivity and land quality in structurally constrained areas.

Future gains will come less from "doing more cycles" and more from optimizing what is grown and where.

Question
Which division has the highest untapped land percentage?

```
SELECT TOP 1
    division,
    ROUND(
        SUM(total_untapped_potential_hectare) * 100.0 /
        NULLIF(SUM(total_agricultural_land_hectare), 0),
    2) AS untapped_pct
FROM dbo.lands
GROUP BY division
ORDER BY untapped_pct DESC;
```

Key Insight:

Mymensingh has the highest untapped land share at ~28%, indicating the largest relative efficiency gap among divisions.

This suggests targeted intervention in this region could yield disproportionately higher gains.

Question
Which types of land contribute most to Bangladesh's untapped agricultural potential?

```
SELECT
    'Fallow Land' AS land_type,
    SUM(current_fallow_land_hectare) AS total_hectare
FROM dbo.lands

UNION ALL

SELECT
    'Char Land',
    SUM(char_land_hectare)
FROM dbo.lands

UNION ALL

SELECT
    'Hilly Land',
    SUM(hilly_land_hectare)
FROM dbo.lands

UNION ALL

SELECT
    'Saline Land',
    SUM(saline_affected_land_hectare)
FROM dbo.lands

UNION ALL

SELECT
    'Rice Fallow Land',
    SUM(rice_fallow_land_hectare)
FROM dbo.lands

ORDER BY total_hectare DESC;
```

Key Insight:

Rice-fallow land dominates untapped potential at ~802K hectares, followed by saline land (~396K hectares), showing that most opportunity lies within already cultivable but underutilized areas.

This suggests Bangladesh can achieve quick gains by improving cropping cycles and land use efficiency, rather than expanding into difficult terrains.

Policy focus should prioritize rice-fallow optimization first, while treating saline and hilly regions as long-term investment zones.

SQL Queries

Question
Which divisions have the highest untapped agricultural land potential in Bangladesh?

```
SELECT
   division,

   ROUND(SUM(total_agricultural_land_hectare), 0) AS total_land,
   ROUND(SUM(total_untapped_potential_hectare), 0) AS untapped_land,

   ROUND(
      SUM(total_untapped_potential_hectare) * 100.0 /
      NULLIF(SUM(total_agricultural_land_hectare), 0),
   2) AS untapped_pct

FROM dbo.lands

GROUP BY division

ORDER BY untapped_land DESC;
```

Key Insight:

Chittagong (~289K ha) and Dhaka (~256K ha) lead in total untapped land, but Barisal (~26.8%) and Mymensingh (~28.4%) show the highest underutilization rates.

This reveals a clear split between scale and inefficiency — large divisions drive total opportunity, while smaller ones suffer deeper structural underuse.

Policy should target high-volume regions for impact, while focusing on high-percentage regions to improve land efficiency.

Question
Which districts have the highest untapped agricultural potential in both absolute terms and as a share of total land?

```
SELECT
   district_name,
   division,

   ROUND(total_agricultural_land_hectare, 0) AS total_land,
   ROUND(total_untapped_potential_hectare, 0) AS untapped_land,

   ROUND(
      (total_untapped_potential_hectare * 100.0)
      / NULLIF(total_agricultural_land_hectare, 0),
   2) AS untapped_pct

FROM dbo.lands

ORDER BY untapped_land DESC;
```

Key Insight:

Out of Bangladesh's total agricultural land, a significant share remains underutilized, with opportunity highly concentrated in a few districts.

Mymensingh (~72K ha), Bhola (~58K ha), and Noakhali (~52K ha) lead both in absolute unused land and high underutilization rates (~25–32%), making them the most impactful targets.

This shows that the problem is not evenly distributed — targeted regional intervention will deliver far greater returns than uniform national policies.

SQL Queries

Question
Which district has the lowest cropping intensity?

```
SELECT TOP 1
   district_name,
   ROUND(cropping_intensity_pct, 2) AS cropping_intensity_pct
FROM dbo.lands
ORDER BY cropping_intensity_pct ASC;
```

Key Insight:

Bandarban shows the lowest cropping intensity at 195%, but the gap is marginal and largely driven by hilly terrain constraints.

This indicates that productivity improvements here will require land-specific strategies, not simply increasing cropping cycles.

Question
Which district offers the highest immediate return for agricultural investment?

```
SELECT TOP 1
   district_name,
   ROUND(total_untapped_potential_hectare, 0) AS untapped_land,
   ROUND(cropping_intensity_pct, 2) AS cropping_intensity_pct
FROM dbo.lands
WHERE cropping_intensity_pct >= 195
ORDER BY total_untapped_potential_hectare DESC;
```

Key Insight:

Mymensingh stands out as the most actionable investment target, with over 72K hectares of untapped land despite already high cropping intensity (~206%).

This indicates a rare combination of scale and efficiency, where targeted investment can deliver immediate and scalable production gains.

Question
Which districts should be prioritized for agricultural investment based on untapped land, efficiency, and feasibility?

```
SELECT
   district_name,
   division,
   primary_area_type,

   ROUND(total_untapped_potential_hectare, 0) AS untapped_land,
   ROUND(cropping_intensity_pct, 2) AS cropping_intensity_pct,

   -- Feasibility factor (penalize difficult terrains)
   CASE
      WHEN primary_area_type IN ('hilly', 'char') THEN 0.6
      WHEN primary_area_type = 'saline' THEN 0.7
      ELSE 1.0
   END AS feasibility_factor,

   -- Final Priority Score
   ROUND(
      (total_untapped_potential_hectare / 1000.0) *
      (cropping_intensity_pct / 200.0) *
      CASE
         WHEN primary_area_type IN ('hilly', 'char') THEN 0.6
         WHEN primary_area_type = 'saline' THEN 0.7
         ELSE 1.0
      END,
   2) AS priority_score,

   RANK() OVER (
      ORDER BY
      (total_untapped_potential_hectare / 1000.0) *
      (cropping_intensity_pct / 200.0) *
      CASE
         WHEN primary_area_type IN ('hilly', 'char') THEN 0.6
         WHEN primary_area_type = 'saline' THEN 0.7
         ELSE 1.0
      END DESC
   ) AS priority_rank

FROM dbo.lands

ORDER BY priority_score DESC;
```

Key Insight:

Mymensingh clearly ranks as the top priority district, combining the largest untapped land (~72K hectares) with already high farming intensity, indicating strong readiness for rapid gains.

The top-ranked districts are not the most extreme or constrained regions, but those where scale, efficiency, and feasibility align — such as Noakhali, Kishoreganj, and Patuakhali.

This confirms that the fastest impact will come from targeting high-readiness zones, rather than investing heavily in structurally difficult areas.

KrishiGrid Bangladesh

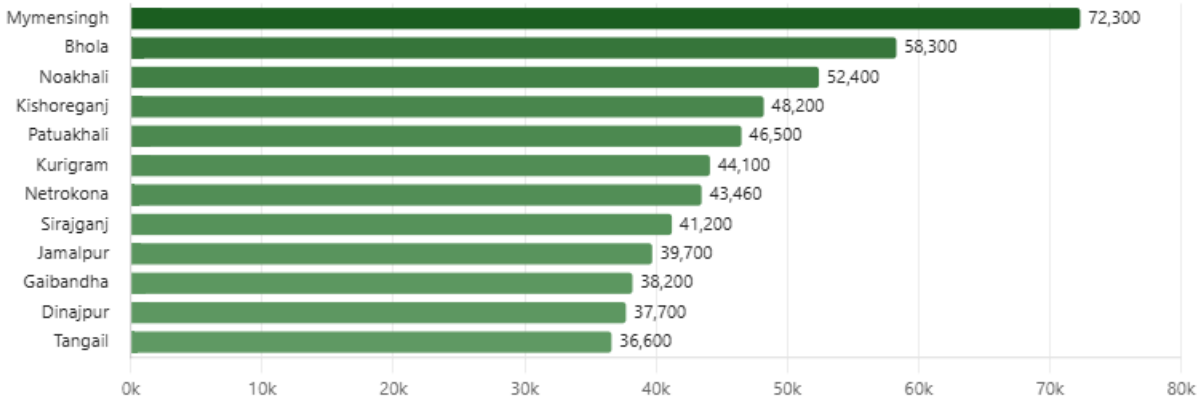
Total Agricultural Land
8,287,000
Hectares

Total Untapped Potential
1,690,360
Hectares

Rice-Fallow Land
820,260
Hectares

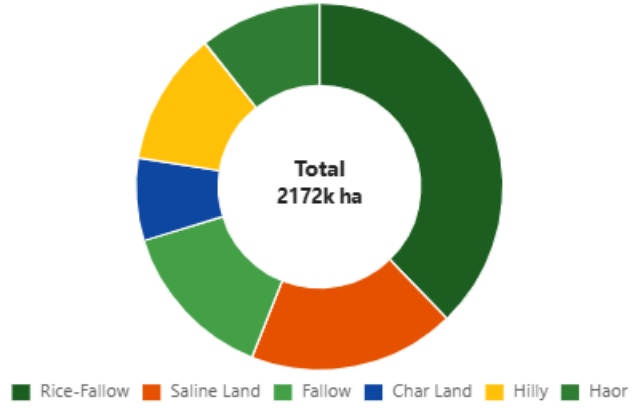
Annual Farmland Lost
19,680
Hectares / Year

Top 12 Districts by Untapped Land (Hectares)



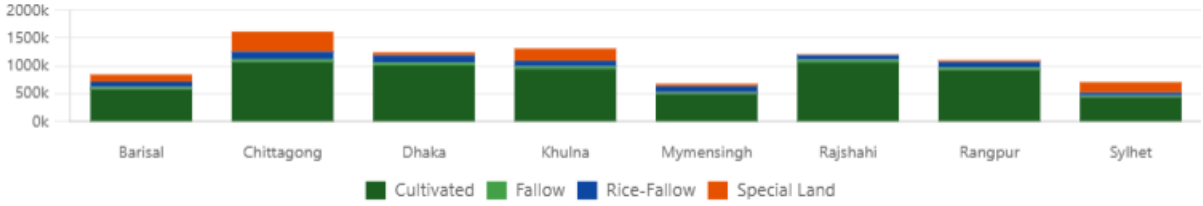
Mymensingh, Bhola, and Noakhali lead in unused land. Targeted investment here delivers far greater returns than uniform national policies.

Types of Untapped Land — National Total



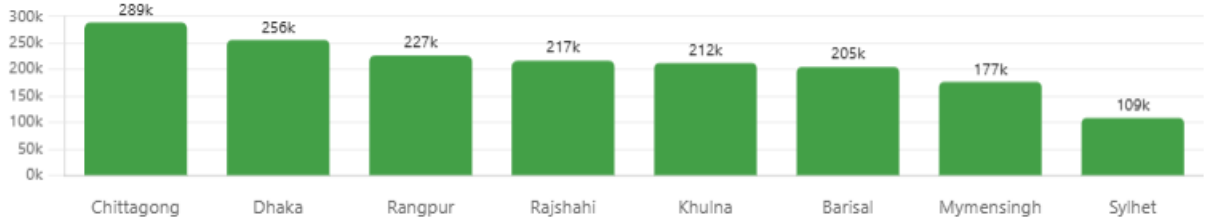
Rice-fallow land is the largest and easiest category to activate — it only requires a cropping pattern change from 2 crops to 3 crops per year.

Land Use Breakdown by Division



Barisal and Mymensingh show the highest underutilization rates relative to total land. These divisions have the most room for immediate improvement.

Untapped Land by Division (Hectares)



Chittagong and Dhaka divisions have large absolute untapped land but high urbanization pressure. Policy must protect remaining farmland urgently.

KrishiGrid Bangladesh

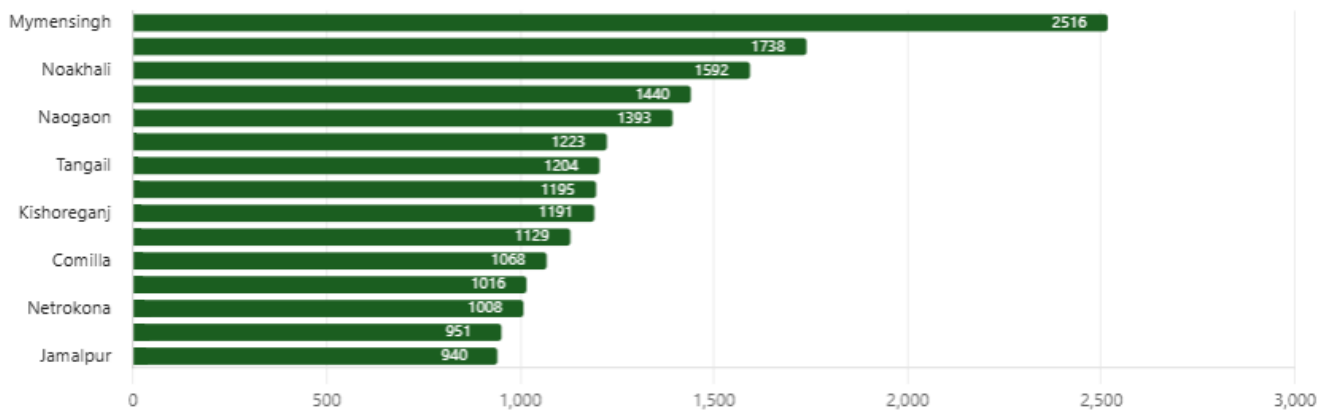
National Avg Cropping Intensity
202.2
%

Lowest Intensity District
Bandarban
195.0% Cropping Intensity

Highest Priority District
Mymensingh
Score: 2,515.9

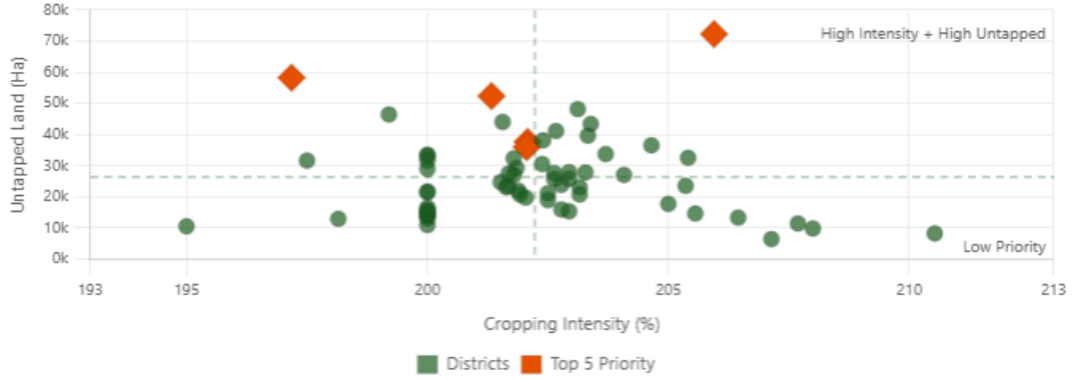
10-Year Farmland Loss Risk
196,800
Hectares

District Investment Priority Score (Top 15)



Priority score combines untapped land size, cropping efficiency gap, and farmer density. Mymensingh ranks highest — scale, efficiency, and feasibility all align.

Cropping Intensity vs Untapped Land

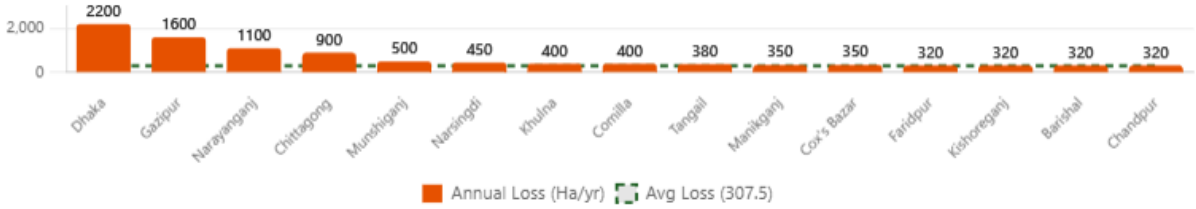


Districts in the top-right zone have both high intensity and high untapped potential — the most actionable targets for KrishiGrid ward hub deployment.

Cropping Intensity by District (Bottom 20)

Districts below 195% cropping intensity are significantly underperforming. Hilly and char districts dominate this category and need terrain-specific solutions.

Annual Farmland Loss by Top 15 Districts (Ha/Year)



Urban-adjacent districts are losing farmland fastest. Without protective legislation, these losses will permanently reduce Bangladesh's food production capacity.

KEY INSIGHTS

Part 2/8

1 Bangladesh has ~1.69 million hectares of underutilized agricultural land (~20% of total), despite already high cropping intensity (~202%).

2 Rice-fallow land alone contributes ~800K hectares, followed by saline land (~396K hectares), showing that most opportunity lies within cultivable but underutilized land — not new expansion.

3 Untapped potential is unevenly distributed — divisions like Mymensingh (~28%) and Barisal (~26%) show the highest relative inefficiency, while Chittagong and Dhaka drive total volume.

4 At the district level, Mymensingh (~72K ha), Bhola (~58K ha), and Noakhali (~52K ha) combine both scale and high underutilization, making them the most impactful intervention zones.

5 Cropping intensity is already near saturation (~200%), meaning future gains will come from improving land use efficiency and crop optimization — not simply increasing cropping cycles.

COMING NEXT

PART 3 of 8: FROM LAND TO FOOD PRODUCTION

We translate land potential into actual crop output — estimating how much import dependency Bangladesh can realistically reduce through targeted cultivation